Project: Worked examples
Title: Mechanism kinematics: four-bar, slider-crank, cam
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1. Scope and method

A single-degree-of-freedom planar mechanism driven at constant speed: a four-bar linkage or an offset slider-crank (both circuits, position, velocity and acceleration over a sweep of the input angle, Grashof class, transmission angle against a minimum, toggle and dead-centre positions, loop-closure residual, input torque by virtual work), or a cam with a translating follower. Rigid links, no clearance, no friction.

WHAT IT CALCULATES	RELATION
Four-bar position	$a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} = d + c\cdot e^{j\theta_4}$; $\ell = \lvert A - O_4\rvert$, $\cos\gamma = (c^2 + \ell^2 - b^2)/(2c\ell)$, $\theta_4 = \varphi \mp \gamma$ (φ the direction $O_4\to A$; open: −, crossed: +); θ_3 from B − A
Four-bar velocity and acceleration	differentiated loop, 2×2 system with det = $b\cdot c\cdot\sin(\theta_3 - \theta_4)$; $\omega_3 = \omega_2\cdot a\cdot\sin(\theta_4 - \theta_2)/(b\cdot\sin(\theta_3 - \theta_4))$, $\omega_4 = \omega_2\cdot a\cdot\sin(\theta_3 - \theta_2)/(c\cdot\sin(\theta_3 - \theta_4))$; α from the second derivative at constant ω_2
Grashof class and transmission angle	$s + l < p + q$ (crank-rocker, double-crank or double-rocker by the shortest link), = at the change point, > non-Grashof; transmission angle $\cos\mu = (b^2 + c^2 - \ell^2)/(2bc)$ folded to 0 … 90°; toggle where $\ell = b + c$ or $\lvert b - c\rvert$
Slider-crank position	$x = r\cdot\cos\theta + \sigma\cdot\sqrt{(l^2 - (r\cdot\sin\theta - e)^2)}$ (σ = +1 open, −1 crossed); $dx/d\theta = -r\cdot\sin\theta - r\cdot\cos\theta\cdot(r\cdot\sin\theta - e)/(x - r\cdot\cos\theta)$; dead centres where crank and rod are collinear; μ = 90° − |rod angle to the slide line|
Input torque by virtual work	$T_{in}\cdot\omega_{in} = \Sigma F\cdot v$; four-bar $T_{in} = (T_L + J\cdot\alpha_4)\cdot\omega_4/\omega_2$; slider $T_{in} = (m\cdot a + F_L)\cdot dx/d\theta$; RMS torque and mean power over the sweep by Simpson's rule
Mechanism forces	four-bar tangential pin force $P = (T_L + J\cdot\alpha_4)/c$, coupler (and crank-pin) force P/sin μ; slider slide-direction force $F_x = m\cdot a + F_L$, rod force $F_x/\cos\theta_3$, guide reaction $F_x\cdot\tan\theta_3$; cam contact normal F/cos φ
Assembly limits	four-bar toggles at $\cos\theta_2 = (d^2 + a^2 - \ell^2)/(2ad)$, $\ell = b + c$ or $\lvert b - c\rvert$; slider $\sin\theta = (e \pm l)/r$. The sweep is cut at them and, driven at constant speed into one, the unbounded peaks are not reported
Loop closure	$\lvert a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} - c\cdot e^{j\theta_4} - d\rvert$ (slider: $\lvert r\cdot e^{j\theta} + l\cdot e^{j\theta_3} - (x + j\cdot e)\rvert$) asserted below 1e-9 mm at every step
Cam, per segment with $x = (\theta - \theta_0)/\beta$	rise $s = s_0 + h\cdot f(x)$, return $s = s_0 - h\cdot f(x)$; cycloidal $f = x - \sin(2\pi x)/(2\pi)$; 3-4-5 $f = 10x^3 - 15x^4 + 6x^5$; 4-5-6-7 $f = 35x^4 - 84x^5 + 70x^6 - 20x^7$; modified trapezoid from its acceleration pieces ($C_v = 2$, $C_a = 8\pi/(\pi + 2)$, $C_j = 32\pi^2/(\pi + 2)$)

WHAT IT CALCULATES	RELATION
Cam SVAJ at constant ω	$v = \omega \cdot s'$, $a = \omega^2 \cdot s''$, $j = \omega^3 \cdot s'''$ ($' = d/d\theta$); peak coefficients $C_v = \max f'$, $C_a = \max \lvert f'' \rvert$, $C_j = \max \lvert f''' \rvert$, so $v_{pk} = C_v \cdot h \cdot \omega / \beta$
Pressure angle (roller, offset e, $R_p = R_b + R_r$)	$\varphi = \arctan[(s' - e)/(\sqrt{(R_p^2 - e^2)} + s)]$, which for $e = 0$ is $\arctan[s'/(R_p + s)]$; the base circle that holds $\varphi \le \varphi_{max}$ has $R_{p,req} = \sqrt{(s_o,req^2 + e^2)}$ with $s_o,req = \max(\lvert s' - e \rvert / \tan \varphi_{max} - s)$
Radius of curvature of the pitch curve $P(\theta) = Rot(-\theta) \cdot (e, \sqrt{(R_p^2 - e^2)} + s)$	$\rho = \lvert P' \rvert^3 / (P' \times P'')$, convex where positive; undercut where $\rho_{min} \le R_r$ (FAIL), none where the curve is nowhere convex; flat-faced follower: signed profile radius $\rho = R_b + s + s''$, the minimum of which must stay above zero (FAIL at or below) and the face must reach $\lvert s' \rvert$; the cam turns counter-clockwise with the follower axis at +x offset e
Cam follower force $F = m \cdot a + k \cdot s + F_{pre} + F_{ext}$ (the follower leaves the cam where $F < 0$); input torque by virtual work $T = F \cdot s'$ ($T \cdot \omega = F \cdot v$)	

Basis: Position, velocity and acceleration analysis; transmission angle; Grashof condition[1](Ch. 4, 6, 7); Loop-closure position analysis; velocity and acceleration of planar linkages[2](Ch. 4–6).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of mechanism	Four-bar		spec: Carton line PKG-LINE-7 §3.2 tucker station rev C	any
circuit	Assembly circuit (four-bar, slider-crank): open is $\theta_4 = \varphi - \gamma$ (slider on the +x side of the crank pin), crossed the mirror branch	Open		drawing: PKG-TUCK-02 rev B	any
L1	Ground link d, O_2 to O_4 (four-bar)	150	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L2	Crank length a (four-bar) or r (slider-crank)	45	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L3	Coupler length b (four-bar) or connecting-rod length l (slider-crank)	140	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L4	Rocker length c (four-bar)	100	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
θ_{start}	Input angle at the first row of the table	0	°	spec: Carton line PKG-LINE-7 §3.2: crank zero at the home position rev C	-720 … 720 °

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
θ_{range}	Swept range of the input angle (360° for a full turn)	360	°	spec: Carton line PKG-LINE-7 §3.2: one cycle rev C	1 … 720 °
steps	Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)	72		assumption: 5° table step.	4 … 720
n_{in}	Input speed (constant)	60	rpm	spec: Carton line PKG-LINE-7 §3.2: 60 cartons/min rev C	0.1 … 30000 rpm
μ_{req}	Minimum transmission angle required (linkages)	40	°	assumption: Default minimum transmission angle; set it from your design rule.	0 … 90 °
$load_{dir}$	External load: constant direction, or always opposing the output motion	Opposes motion		assumption: The flap resists the blade in whichever direction it moves.	any
T_{load}	External load torque on the output link (four-bar). Positive acts in the $-\theta_4$ direction, resisting positive ω_4; with 'Opposes motion' its sign is applied against the actual direction of ω_4 (a negative value then assists)	4	N·m	assumption: Example flap-folding torque at the rocker shaft; measure it on the station.	-10000 … 10000 N·m
J_{out}	Moment of inertia of the output link about its pivot (four-bar)	0.012	kg·m²	assumption: Example: rocker arm and blade about the rocker pivot; take it from the CAD mass properties.	0 … 1000 kg·m²

3. Assumptions

A1. Rigid links, a planar mechanism with one degree of freedom, no joint clearance and no friction; the input turns at a constant speed. (Scope limits of the model.)
A2. The position is the closed-form solution of the vector loop; the open and crossed circuits are the two branches ($\theta_4 = \varphi - \gamma$ and $\varphi + \gamma$ for the four-bar, $\sigma = +1$ and -1 for the slider) and the sweep stays on the chosen branch; for a change-point linkage (parallelogram, deltoid) the circuit is named at a reference angle that does not depend on the sweep (the middle of the first interval between change points in 0 … 2π, π/2 when there are none), so the label at a crank angle means the same motion on every turn; only the true change points (all four links collinear, $\theta_2 = 0$ or π) flip the branch, and it is followed by continuity through them inside the sweep. A linkage with one change point per turn (a kite) repeats every 720° and a sweep of more than one turn warns. Angles at which the linkage cannot be assembled, or at a toggle where the velocity system is singular, are left out of the table and counted. (Norton, Design of Machinery, position analysis of the four-bar and the offset slider-crank.)

A3. Input torque is by virtual work from the output load and the inertia of the output element only (four-bar: the output link's moment of inertia about its pivot; slider: the slider mass); the crank and the coupler or connecting rod are massless. (No link-inertia or gravity terms: add the link masses to the output element for a first allowance.)
A4. The peaks (speeds, accelerations, torque, transmission angle, swing) are those of the continuous motion over the swept range, found by a dense scan refined by golden section; the printed table (steps N, up to 3600) only samples it. (Peaks do not depend on the table's step count.)
A5. Cam: a translating follower on a cam turning at constant speed; the follower mass, a linear return spring (rate and preload at the base circle, for example linked from M7) and a constant external load make the force the cam must apply, $F = m{\cdot}a + k{\cdot}s + F_{pre} + F_{ext}$; contact is lost where it falls below zero. The segment table must add to 360° and close; pressure angle and undercut are checked for the roller follower ($R_p = R_b + R_r$), the positive profile radius for the flat-faced one. (Norton, Design of Machinery, cam design: the four classical laws, pressure angle and radius of curvature.)
A6. Follower force (four-bar: the tangential force at the output pin, $(T_L + J{\cdot}\alpha_4)/c$; slider: the slide-direction force $m{\cdot}a + F_L$; cam: the axial force) is the force the mechanism must apply to the output element. The force in the coupler ($F_{pin,pk}$) is the tangential force over sin μ; in the connecting rod it is the slide-direction force over cos θ_3, and the guide reaction is that force times tan θ_3; the roller contact normal force is F/cos φ. Crank, coupler and rod are massless two-force members. (Rigid body, frictionless.)
A7. When the sweep meets an assembly limit (a toggle of a four-bar, the rod perpendicular to the slide line of a slider-crank) the sweep is cut there and, because the speed, acceleration, torque and forces at constant input speed are unbounded at the limit, they are not reported; the limit angles are listed. A change-point four-bar (parallelogram, deltoid) is continued through its singular position on the same branch by continuity (γ changes sign there), but its speeds and torques next to the singular position are not reported. (Closed-form limit angles: $\ell = b + c$ or $|b - c|$ (four-bar), $|r{\cdot}\sin\theta - e| = l$ (slider).)
A8. Into F1 and E1: $output_{motion}$ is a kinematic proxy at constant crank speed, not the cylinder speed of an actuated linkage. For E1 use the peak $input_{torque}$ for the peak check and $T_{rms,in}$ with n_{in} for the thermal sizing; the mean power is P_{mean}. (Used by the actuated-linkage and mechanism-review packs.)
A9. steps (Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)) is taken as 72. (5° table step.)
A10. μ_{req} (Minimum transmission angle required (linkages)) is taken as 40 deg. (Default minimum transmission angle; set it from your design rule.)
A11. $load_{dir}$ (External load: constant direction, or always opposing the output motion) is taken as Opposes motion. (The flap resists the blade in whichever direction it moves.)
A12. T_{load} (External load torque on the output link (four-bar). Positive acts in the $-\theta_4$ direction, resisting positive ω_4; with 'Opposes motion' its sign is applied against the actual direction of ω_4 (a negative value then assists)) is taken as 4 N*m. (Example flap-folding torque at the rocker shaft; measure it on the station.)
A13. J_{out} (Moment of inertia of the output link about its pivot (four-bar)) is taken as 0.012 kg*m^2. (Example: rocker arm and blade about the rocker pivot; take it from the CAD mass properties.)

4. Working

[10] $f_{in} = n_{in} / 60$
= rpm to Hz
= 1 Hz
Input (crank) frequency, for the spring surge check of M7 (mech.f_in)

[16] $S_{plus,L} = s + l$ [3](criterion)
= shortest + longest link
= 195 mm
Shortest plus longest link

[17] $P_{plus,Q} = p + q$ [3](criterion)
= the other two links
= 240 mm
Sum of the other two links

[18] $theta2_{sw}$ = θ_2 [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg
Table: crank angle
[19] $theta3_{sw}$ = θ_3 = atan2(B - A) [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [45.45, 43.305, 41.183, 39.119, 37.143, 35.277, 33.533, 31.922, 30.446, 29.103, 27.892, 26.
items)] deg
Table: coupler angle
[20] $theta4_{sw}$ = θ_4 = φ ▯ γ [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [93.891, 91.886, 90.182, 88.806, 87.774, 87.09, 86.749, 86.738, 87.038, 87.625, 88.475, 89.
items)] deg
Table: rocker angle
[21] $omega3_{sw}$ = ω_3 [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [-2.6928, -2.6892, -2.6368, -2.5435, -2.4183, -2.2706, -2.1088, -1.9402, -1.7705, -1.6037,
-1.2886, … (73 items)] rad/s
Table: coupler angular velocity
[22] $omega4_{sw}$ = ω_4 [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [-2.6928, -2.3371, -1.9398, -1.5152, -1.0781, -0.64174, -0.21757, 0.18575, 0.56219, 0.90807
1.5026, … (73 items)] rad/s
Table: rocker angular velocity
[23] $alpha3_{sw}$ = α_3 [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [-1.6439, 2.0963, 5.3499, 7.9793, 9.9336, 11.238, 11.97, 12.238, 12.157, 11.832, 11.354, 10
items)] rad/s²
Table: coupler angular acceleration
[24] $alpha4_{sw}$ = α_4 [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [23.794, 27.273, 29.765, 31.194, 31.596, 31.098, 29.878, 28.129, 26.035, 23.751, 21.398, 19
items)] rad/s²
Table: rocker angular acceleration
[25] μ_{sw} = μ [4](cosines))
= 73 rows, θ_2 from 0° to 360°
= [48.441, 48.581, 48.999, 49.687, 50.631, 51.814, 53.216, 54.816, 56.592, 58.522, 60.583, 62
items)] deg
Table: transmission angle
[26] $T_{in,sw}$ = $(T_L + J \cdot \alpha_4) \cdot \omega_4 / \omega_2$ [5](work)
= 73 rows, θ_2 from 0° to 360°
= [1.5919, 1.3661, 1.1246, 0.87436, 0.62126, 0.37043, 0.1261, 0.12823, 0.38586, 0.61929, 0.82
… (73 items)] N·m
Table: input torque by virtual work
[27] $theta3_0$ = θ_3 [4](cosines))
= at the first input angle
= 45.45 deg
Coupler angle at the first input angle
[28] $theta4_0$ = θ_4 [4](cosines))
= at the first input angle
= 93.891 deg
Rocker angle at the first input angle

[29] $\omega_{30} = \omega_3$ [4](cosines))
= at the first input angle
= -2.6928 rad/s
Coupler angular velocity at the first input angle
[30] $\omega_{40} = \omega_4$ [4](cosines))
= at the first input angle
= -2.6928 rad/s
Rocker angular velocity at the first input angle
[31] $\alpha_{30} = \alpha_3$ [4](cosines))
= at the first input angle
= -1.6439 rad/s²
Coupler angular acceleration at the first input angle
[32] $\alpha_{40} = \alpha_4$ [4](cosines))
= at the first input angle
= 23.794 rad/s²
Rocker angular acceleration at the first input angle
[33] $\mu_0 = \mu$ [4](cosines))
= at the first input angle
= 48.441 deg
Transmission angle at the first input angle
[34] $T_{in,0} = (T_L + J \cdot \alpha_4) \cdot \omega_4 / \omega_2$ [5](work)
= at the first input angle
= 1.5919 N·m
Input torque at the first input angle
[35] $\theta_{4,alt}$ = θ_4 on the other circuit [4](cosines))
= mirror branch, at the first input angle
= 266.11 deg
Rocker angle on the other circuit at the first input angle
[36] $\theta_{4,min}$ = min θ_4 [4](cosines))
= unwrapped along the sweep
= 86.704 deg
Smallest rocker angle
[37] $\theta_{4,max}$ = max θ_4 [4](cosines))
= unwrapped along the sweep
= 141.49 deg
Largest rocker angle
[38] swing = max θ_4 - min θ_4 [4](cosines))
= over the sweep
= 54.786 deg
Rocker swing over the sweep
[39] μ_{min} = min μ [4](cosines))
= dense scan, refined
= 48.441 deg
Smallest transmission angle over the sweep
[40] res_{max} = max |loop residual| [4](cosines))
= 73 table rows and 1441 scan angles
= 9.3095e-14 mm
Largest loop-closure residual
[41] $res_{vel,max}$ = max |velocity loop| / its terms [4](cosines))
= 73 table rows
= 2.1412e-16
Largest velocity-loop residual, relative to the size of its terms

[42] $\text{omega3}_{pk} = \max |\omega_3|$ [4](cosines))
= dense scan, refined
= 2.6976 rad/s
Peak coupler angular velocity
[43] $\omega_{out,pk} = \max |\omega_4|$ [4](cosines))
= dense scan, refined
= 3.5435 rad/s
Peak output (rocker) angular velocity
[44] $\text{alpha3}_{pk} = \max |\alpha_3|$ [4](cosines))
= dense scan, refined
= 22.724 rad/s²
Peak coupler angular acceleration
[45] $\alpha_{out,pk} = \max |\alpha_4|$ [4](cosines))
= dense scan, refined
= 31.599 rad/s²
Peak output (rocker) angular acceleration
[46] $\text{input}_{torque} = \max |T_{in}|$ [5](work)
= dense scan, refined
= 2.2654 N·m
Peak input (crank) torque by virtual work (mech.input_torque → E1 peak torque)
[47] $\text{output}_{motion} = c \cdot \max |\omega_4|$ [4](cosines))
= output pin at the rocker length
= 0.35435 m/s
Peak speed of the output pin at the rocker length c (mech.output_motion; a kinematic proxy for F1 only if the cylinder acts perpendicular to the rocker at that radius; the angular velocity is $\omega_{out,pk}$)
[48] $\text{peak}_{accel} = \max |\alpha_{pin}|$ [4](cosines))
= $\sqrt{(\alpha_4^2 + \omega_4^4)} \cdot c$
= 3.1625 m/s²
Peak acceleration of the output pin (mech.peak_accel)
[49] $\text{follower}_{force} = \max |(T_L + J \cdot \alpha_4)/c|$ [5](work)
= tangential force at the output pin
= 43.476 N
Peak force perpendicular to the rocker at the output pin, the drive force of the output (mech.follower_force)
[50] $F_{pin,pk} = \max |(T_L + J \cdot \alpha_4)/(c \cdot \sin \mu)|$ [5](work)
= force in the coupler = the crank-pin force
= 53.69 N
Peak force in the coupler, which is also the force at the crank pin and the pin loads on the bearings (mech.F_pin_pk → M4, M5)
[51] $T_{rms,in} = \sqrt{(\text{mean } T_{in}^2)}$ [5](work)
= Simpson over the scan
= 1.3744 N·m
RMS input torque over the swept range (mech.T_rms_in → E1 thermal sizing)
[52] $P_{mean} = \omega \cdot \text{mean}(T_{in})$ [5](work)
= Simpson over the scan
= 7.6496 W
Mean input power over the swept range (negative: the mechanism returns energy) (mech.P_mean)
[53] $\mu_{mu} = \mu_{req} / \mu_{min}$ [6](angle)
= 40° / 48.441°
= 0.82575
Transmission angle: required minimum against the smallest over the sweep

5. Results

SYMBOL	VALUE	DESCRIPTION
n_{in}	60 rpm	Input (crank) speed, passed through for E1's load speed (mech.n_in)

SYMBOL	VALUE	DESCRIPTION
f_{in}	1 Hz	Input (crank) frequency, for the spring surge check of M7 (mech.f_in)
grashof	"Grashof crank-rocker, input link shortest"	Grashof classification
$S_{plus,L}$	195 mm	Shortest plus longest link
$P_{plus,Q}$	240 mm	Sum of the other two links
$theta2_{sw}$	[0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg	Table: crank angle
$theta3_{sw}$	[45.45, 43.305, 41.183, 39.119, 37.143, 35.277, 33.533, 31.922, 30.446, 29.103, 27.892, 26.806, … (73 items)] deg	Table: coupler angle
$theta4_{sw}$	[93.891, 91.886, 90.182, 88.806, 87.774, 87.09, 86.749, 86.738, 87.038, 87.625, 88.475, 89.561, … (73 items)] deg	Table: rocker angle
$omega3_{sw}$	[-2.6928, -2.6892, -2.6368, -2.5435, -2.4183, -2.2706, -2.1088, -1.9402, -1.7705, -1.6037, -1.4425, -1.2886, … (73 items)] rad/s	Table: coupler angular velocity
$omega4_{sw}$	[-2.6928, -2.3371, -1.9398, -1.5152, -1.0781, -0.64174, -0.21757, 0.18575, 0.56219, 0.90807, 1.2216, 1.5026, … (73 items)] rad/s	Table: rocker angular velocity
$alpha3_{sw}$	[-1.6439, 2.0963, 5.3499, 7.9793, 9.9336, 11.238, 11.97, 12.238, 12.157, 11.832, 11.354, 10.795, … (73 items)] rad/s²	Table: coupler angular acceleration
$alpha4_{sw}$	[23.794, 27.273, 29.765, 31.194, 31.596, 31.098, 29.878, 28.129, 26.035, 23.751, 21.398, 19.064, … (73 items)] rad/s²	Table: rocker angular acceleration
μ_{sw}	[48.441, 48.581, 48.999, 49.687, 50.631, 51.814, 53.216, 54.816, 56.592, 58.522, 60.583, 62.755, … (73 items)] deg	Table: transmission angle

SYMBOL	VALUE	DESCRIPTION
$T_{in,sw}$	[1.5919, 1.3661, 1.1246, 0.87436, 0.62126, 0.37043, 0.1261, 0.12823, 0.38586, 0.61929, 0.82764, 1.0113, … (73 items)] N·m	Table: input torque by virtual work
$theta3_0$	45.45 deg	Coupler angle at the first input angle
$theta4_0$	93.891 deg	Rocker angle at the first input angle
$omega3_0$	-2.6928 rad/s	Coupler angular velocity at the first input angle
$omega4_0$	-2.6928 rad/s	Rocker angular velocity at the first input angle
$alpha3_0$	-1.6439 rad/s²	Coupler angular acceleration at the first input angle
$alpha4_0$	23.794 rad/s²	Rocker angular acceleration at the first input angle
μ_0	48.441 deg	Transmission angle at the first input angle
$T_{in,0}$	1.5919 N·m	Input torque at the first input angle
$theta4_{alt}$	266.11 deg	Rocker angle on the other circuit at the first input angle
$theta4_{min}$	86.704 deg	Smallest rocker angle
$theta4_{max}$	141.49 deg	Largest rocker angle
swing	54.786 deg	Rocker swing over the sweep
μ_{min}	48.441 deg	Smallest transmission angle over the sweep
res_{max}	9.3095e-14 mm	Largest loop-closure residual
$res_{vel,max}$	2.1412e-16	Largest velocity-loop residual, relative to the size of its terms
$toggle_{angles}$	"none: coupler and rocker are never collinear"	Toggle crank angles (coupler and rocker collinear, μ = 0): the limits of the crank's assemblable range
$limit_{positions}$	"θ_2 = 32.66° with θ_4 = 86.704°; θ_2 = 220.95° with θ_4 = 141.49°"	Crank and coupler collinear (the rocker's dead-centre positions)
$omega3_{pk}$	2.6976 rad/s	Peak coupler angular velocity
$\omega_{out,pk}$	3.5435 rad/s	Peak output (rocker) angular velocity
$alpha3_{pk}$	22.724 rad/s²	Peak coupler angular acceleration
$\alpha_{out,pk}$	31.599 rad/s²	Peak output (rocker) angular acceleration
$input_{torque}$	2.2654 N·m	Peak input (crank) torque by virtual work (mech.input_torque → E1 peak torque)
$output_{motion}$	0.35435 m/s	Peak speed of the output pin at the rocker length c (mech.output_motion; a kinematic proxy for F1 only if the cylinder acts perpendicular to the rocker at that radius; the angular velocity is $\omega_{out,pk}$)
$peak_{accel}$	3.1625 m/s²	Peak acceleration of the output pin (mech.peak_accel)

SYMBOL	VALUE	DESCRIPTION
$follower_{force}$	43.476 N	Peak force perpendicular to the rocker at the output pin, the drive force of the output (mech.follower_force)
$F_{pin,pk}$	53.69 N	Peak force in the coupler, which is also the force at the crank pin and the pin loads on the bearings (mech.F_pin_pk → M4, M5)
$T_{rms,in}$	1.3744 N·m	RMS input torque over the swept range (mech.T_rms_in → E1 thermal sizing)
P_{mean}	7.6496 W	Mean input power over the swept range (negative: the mechanism returns energy) (mech.P_mean)
u_{mu}	0.82575	Required against smallest transmission angle

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Transmission angle μ_{min} at or above the required minimum	u_{mu} = 0.82575	u_{mu} ≤ 1	0.826	[6](angle)
PASS	Loop-closure residual below 1e-9 mm at every step	max residual = 9.3095e-14 mm	< 1e-9 mm	0	[7]
PASS	Velocity-loop residual below 1e-8 of the size of its terms	max residual = 2.1412e-16	< 1e-8	0	[7]

7. Applicability

No warnings.

8. Verification

```
Template mech.mechanism@1.2.1 Content hash d38ef071d1fe242e Engine calc-engine/1.0.0
Known-answer cases passed 16/16 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Norton, Design of Machinery.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
3. Grashof.
4. Vector-loop closed form (Freudenstein / law of.
5. Virtual.
6. Transmission.
7. Self-consistency.